

BEOGRAD, Serbia

WMO: 132740

Lat: 44.7984N

Lon: 20.4648E

Elev: 433

StdP: 14.47

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.5	20.5	10.0	9.3	17.6	14.1	11.4	23.5	20.5	32.0	17.8	34.3	5.8	120	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	94.3	70.6	91.2	70.1	88.2	69.1	73.2	88.4	71.7	86.7	70.3	84.7	5.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.5	106.5	80.2	66.8	100.3	77.9	65.4	95.5	76.4	37.0	88.8	35.6	86.4	34.5	84.9	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	13.8	11.7	DB	12.8	98.9	5.3	3.3	9.0	101.3	5.8	103.2	2.8	105.0	-1.0	107.4	(4)
(5)			WB	11.7	75.9	5.0	1.7	8.1	77.1	5.2	78.1	2.4	79.1	-1.3	80.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.2	35.6	39.3	47.3	56.8	65.0	71.3	74.8	75.1	66.0	56.6	47.4	37.8		
	DBStd	16.02	9.01	9.68	8.73	7.70	7.17	7.20	6.44	6.58	7.34	8.12	8.85	8.54		
	HDD50	1526	451	317	156	29	1	0	0	0	0	31	153	388		
	HDD65	4294	910	721	550	259	93	24	5	6	77	276	530	843		
	CDD50	3777	6	17	71	234	465	639	769	778	479	236	74	9		
	CDD65	1072	0	0	0	14	92	213	310	319	106	17	1	0		
	CDH74	9793	0	0	2	107	688	1991	3064	3093	768	79	1	0		
	CDH80	3825	0	0	0	18	155	749	1303	1362	231	7	0	0		
(14) Wind	WSAvg	5.2	5.5	5.9	5.8	5.6	4.8	4.5	4.4	4.3	4.7	5.2	6.0	5.4		
(15) Precipitation	PrecAvg	26.8	1.9	1.7	2.0	2.2	2.8	3.5	2.6	2.0	2.0	1.9	2.0	2.2		
	PrecMax	43.1	4.4	5.0	5.7	6.2	11.0	8.3	10.3	6.6	5.3	7.3	5.2	7.0		
	PrecMin	18.3	0.2	0.0	0.1	0.2	0.4	0.6	0.1	0.2	0.1	0.0	0.2	0.0		
	PrecStd	5.5	1.1	1.2	1.3	1.0	1.8	1.9	2.1	1.5	1.3	1.5	1.1	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.9	65.4	72.9	82.4	88.1	94.1	98.5	99.5	91.9	80.1	71.5	62.0		
		MCWB	50.2	50.9	54.9	61.6	66.9	70.9	71.0	71.6	68.5	64.2	58.2	52.7		
	2%	DB	55.1	60.7	68.5	76.7	84.1	90.9	94.1	94.9	86.0	76.1	67.1	57.3		
		MCWB	47.3	49.5	53.0	59.2	65.2	70.5	70.9	70.4	66.5	62.1	56.2	49.7		
	5%	DB	51.8	56.8	64.9	72.8	81.2	87.8	90.7	91.3	81.7	72.5	63.5	53.5		
		MCWB	45.8	47.7	51.7	57.0	64.3	69.5	70.4	69.8	65.2	60.6	54.4	47.3		
	10%	DB	48.3	53.0	60.7	69.3	77.9	84.6	87.3	87.6	78.1	68.8	60.0	49.6		
		MCWB	43.4	45.5	49.6	55.3	62.9	68.1	69.1	68.8	63.8	59.1	52.4	44.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.6	52.6	56.7	63.4	69.0	75.2	74.8	75.8	70.1	65.8	59.5	53.6		
		MCDB	58.6	62.6	70.1	79.6	83.4	88.9	90.0	90.9	87.4	77.5	68.7	60.0		
	2%	WB	48.0	50.2	54.1	60.2	66.7	72.4	72.9	72.9	67.8	63.4	57.0	50.2		
		MCDB	53.9	59.4	65.8	73.6	80.4	86.5	88.1	88.5	82.0	73.3	65.5	56.5		
	5%	WB	45.9	48.2	52.2	58.3	65.3	70.7	71.5	71.3	66.3	61.5	55.0	47.7		
		MCDB	51.3	55.8	63.2	70.1	78.6	84.4	86.6	86.6	78.8	70.5	62.1	53.0		
	10%	WB	43.9	46.1	50.2	56.5	63.7	69.1	70.2	69.9	64.7	59.7	53.0	45.0		
		MCDB	48.1	52.5	59.5	67.0	76.1	82.1	84.3	84.3	75.9	67.7	59.5	49.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	11.1	14.1	15.8	16.9	17.0	17.7	18.1	15.9	14.6	11.3	8.6		
		MCDBR	13.8	16.9	20.2	21.1	21.0	21.1	22.1	22.6	21.0	19.3	16.5	13.9		
	5% WB	MCWBR	8.7	9.4	10.0	9.1	8.4	7.8	7.5	7.8	7.7	8.9	8.5	8.9		
		MCDWR	13.0	15.6	18.5	18.9	19.6	19.5	19.8	20.3	19.2	17.9	15.2	13.1		
(40) Clear-Sky Solar Irradiance	taub		0.345	0.371	0.402	0.458	0.446	0.441	0.444	0.454	0.432	0.396	0.359	0.342		
	taud		2.336	2.249	2.216	2.069	2.167	2.217	2.205	2.177	2.206	2.301	2.361	2.355		
	Ebn at Noon		238	252	261	255	263	264	261	253	248	240	231	226		
	Edh at Noon		26	33	39	49	46	44	44	44	39	31	25	23		
(44) All-Sky Solar Radiation	RadAvg		393	655	1046	1439	1753	1947	1972	1748	1225	843	514	330		
	RadStd		54	87	124	146	169	182	150	164	133	93	43	48		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)		+1.22	N/A	N/A	N/A	+0.67	+0.79	-198	-302	+203	+108

Nomenclature: See separate page